



Graphs in Machine Learning

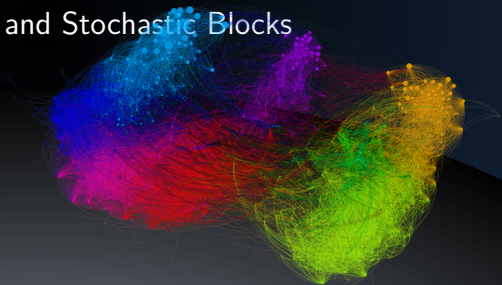
Random Graph Models

Erdős–Rényi, Barabási–Albert, and Stochastic Blocks

Michal Valko

Isara Labs

Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns



The in-between option: random graph models

Erdős-Rényi

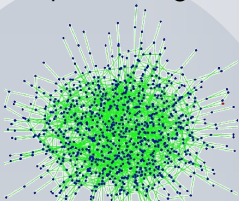
independent edges



The in-between option: random graph models

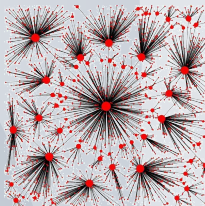
Erdős-Rényi

independent edges



Barabási-Albert

preferential attachment



The in-between option: random graph models

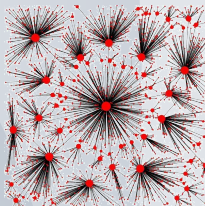
Erdős-Rényi

independent edges



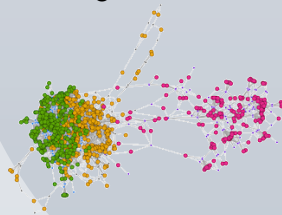
Barabási-Albert

preferential attachment



Stochastic Blocks

modeling communities



The in-between option: random graph models

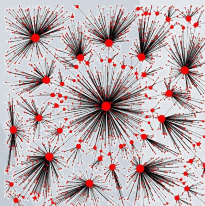
Erdős-Rényi

independent edges



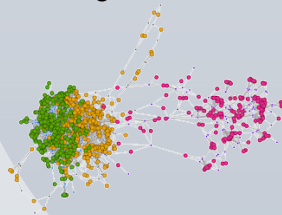
Barabási-Albert

preferential attachment

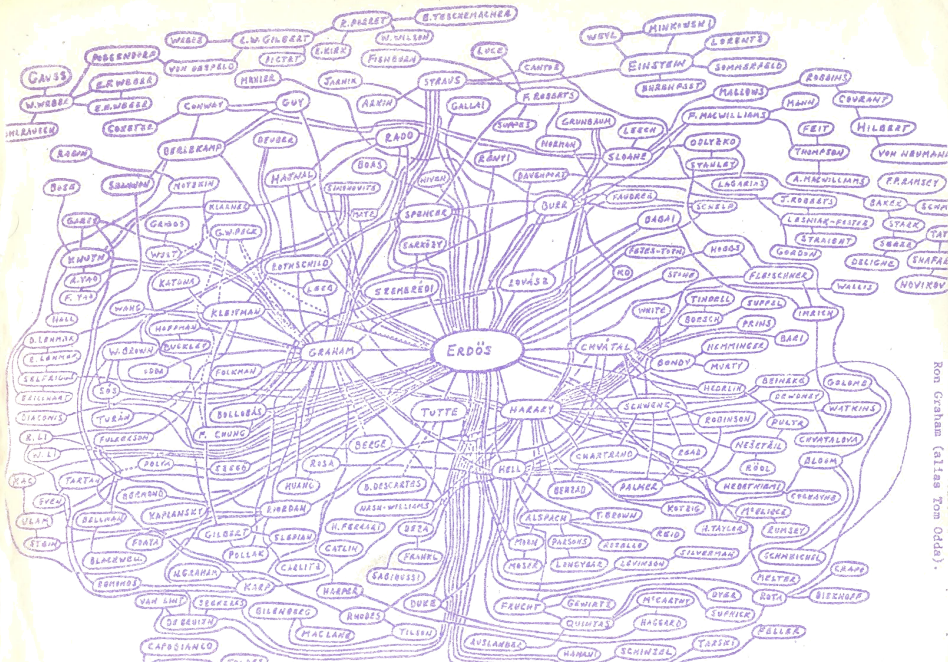


Stochastic Blocks

modeling communities



Watts-Strogatz, Chung-Lu, Fiedler,

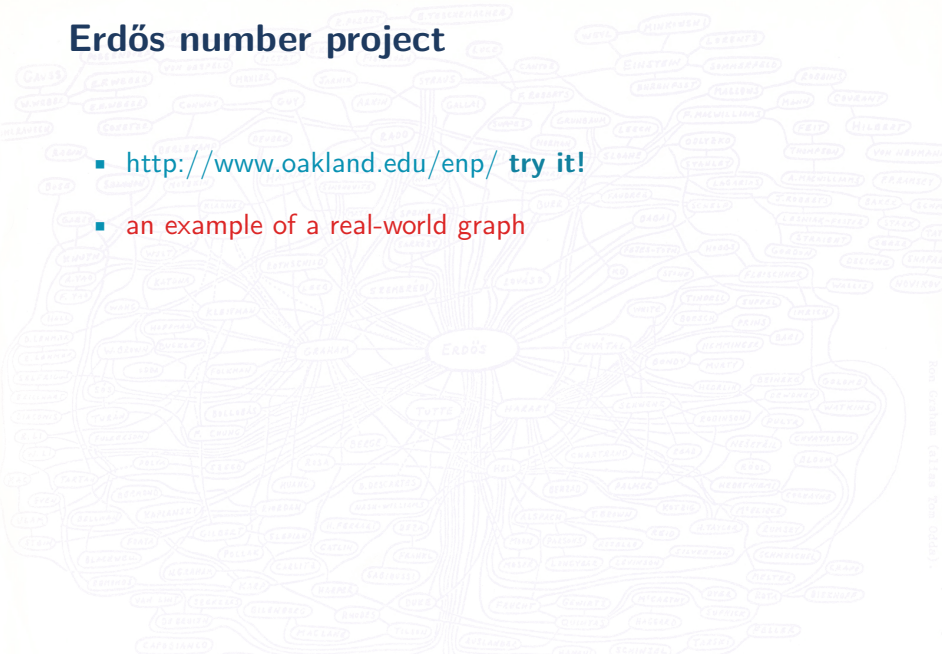


[illegible]

- <http://www.oakland.edu/enp/> **try it!**

Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph



Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph
- 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)

Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph
- 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)
- average degree 3.36

Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph
- 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)
- average degree 3.36
- average distance for the largest component: 7.64

Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph
- 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)
- average degree 3.36
- average distance for the largest component: 7.64
- 6 degrees of separation [Travers & Milgram, 1967]

Erdős number project

- <http://www.oakland.edu/enp/> try it!
- an example of a real-world graph
- 401 000 authors, 676 000 edges ($\ll 401000^2 \rightarrow$ sparse)
- average degree 3.36
- average distance for the largest component: 7.64
- 6 degrees of separation [Travers & Milgram, 1967]
- heavy tail

Michal Valko

`michal.valko@inria.fr`

Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>